

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1-Pseudocode Design

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5-: Pseudocode Design	Assessment:	Assessment Tool 1-: Pseudocode Design
Weightage:	30% of CIA	Total Marks:	100
CO5 Statement:	Analyze and design pseudocode for applying machine learning algorithms to preprocess data, build models, make predictions, perform clustering, and evaluate model performance using appropriate metrics. (BL4)		
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Section / Batch:	CSE AI 2024	Date:	24-08-2026

Code of Conduct

I __Y SRAVAN KUMAR (192472289) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ SRAVAN _____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	
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CO5-AT1: Pseudocode Design

Experiment 1 – Unsupervised Learning

Aim

To design a pseudocode and implement an unsupervised learning process to preprocess unlabelled data, discover patterns, and generate cluster assignments.

Pseudocode

START

Load	unlabelled	dataset
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Remove unnecessary data

Handle	missing	values
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Scale or normalize features

Select an unsupervised learning algorithm

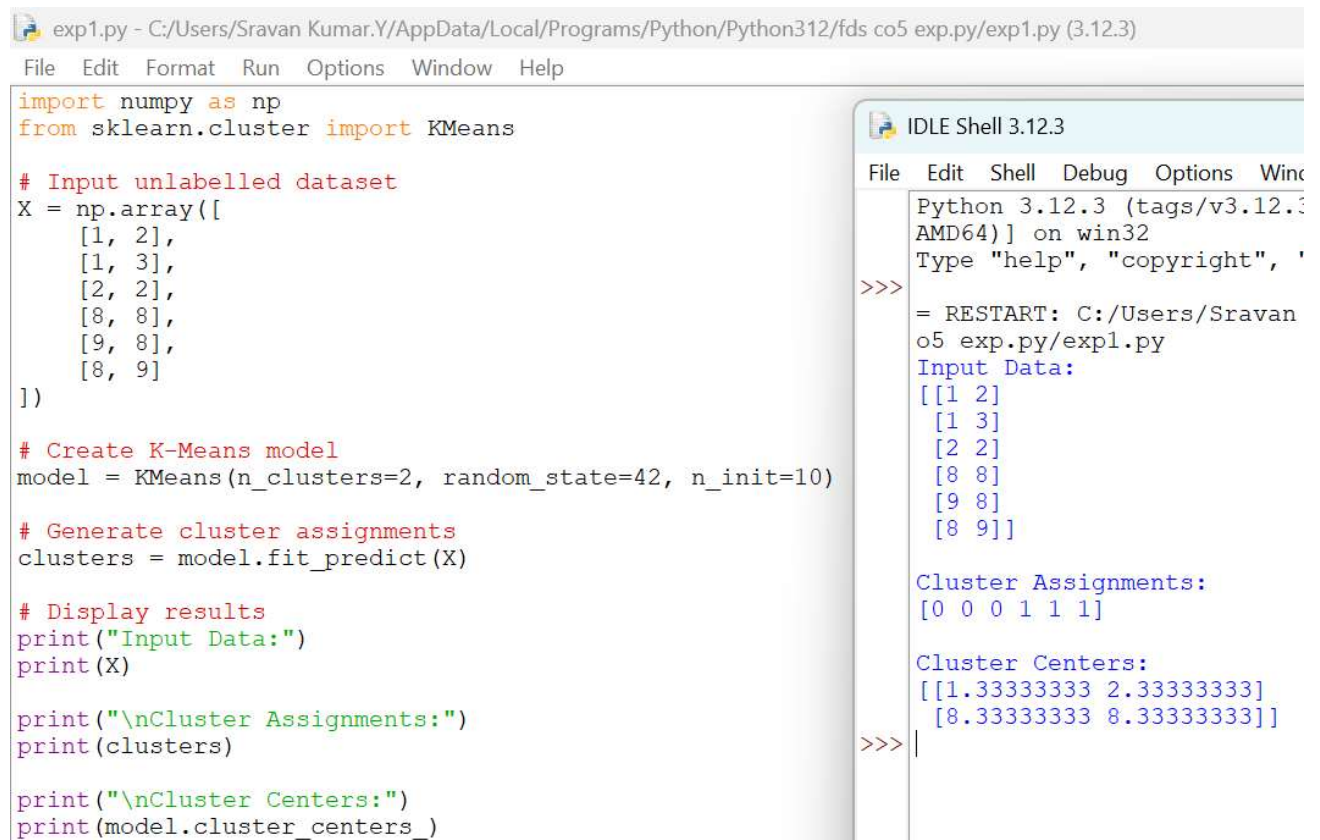
Discover patterns or groups

Generate cluster assignments

Display discovered groups

STOP

Code



```
exp1.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds co5 exp.py/exp1.py (3.12.3)
File Edit Format Run Options Window Help

import numpy as np
from sklearn.cluster import KMeans

# Input unlabelled dataset
X = np.array([
    [1, 2],
    [1, 3],
    [2, 2],
    [8, 8],
    [9, 8],
    [8, 9]
])

# Create K-Means model
model = KMeans(n_clusters=2, random_state=42, n_init=10)

# Generate cluster assignments
clusters = model.fit_predict(X)

# Display results
print("Input Data:")
print(X)

print("\nCluster Assignments:")
print(clusters)

print("\nCluster Centers:")
print(model.cluster_centers_)

IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:66b4f133, Dec 4 2023) on win32
Type "help", "copyright", "credits()" or "quit()" for more.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds co5 exp.py/exp1.py
Input Data:
[[1 2]
 [1 3]
 [2 2]
 [8 8]
 [9 8]
 [8 9]]

Cluster Assignments:
[0 0 0 1 1 1]

Cluster Centers:
[[1.33333333 2.33333333]
 [8.33333333 8.33333333]]
>>>
```

Result

Thus, the unlabelled dataset was successfully processed and grouped into clusters using unsupervised learning.

Experiment 2 – K-Means Clustering

Aim

To design pseudocode and implement K-Means clustering by initializing centroids, assigning observations, updating centroids, and repeating until convergence.

Pseudocode

START

Input dataset

Choose number of clusters K

Initialize K random centroids

REPEAT

 Assign each observation to nearest centroid

 Calculate new centroid for each cluster

 Update centroids

UNTIL centroids do not change

Display final clusters

STOP

Code

```
fds co5 exp2.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Pyth
File Edit Format Run Options Window Help
import numpy as np
from sklearn.cluster import KMeans

X = np.array([
    [1, 2],
    [1, 3],
    [2, 2],
    [8, 8],
    [9, 8],
    [8, 9]
])

model = KMeans(n_clusters=2, random_state=42, n_init=10)

model.fit(X)

print("Cluster Labels:")
print(model.labels_)

print("Centroids:")
print(model.cluster_centers_)

IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:0
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for
.>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Progr
o5 exp2.py
Cluster Labels:
[0 0 0 1 1 1]
Centroids:
[[1.33333333 2.33333333]
 [8.33333333 8.33333333]]
.>>|
```

Result

Thus, the observations were successfully grouped into two clusters using the K-Means clustering algorithm.

Experiment 3 – K-Means Customer Segmentation

Aim

To design pseudocode and implement K-Means clustering to group customers into three clusters based on annual income and spending score.

Pseudocode

START

Load customer dataset

Select:

X_1 = Annual Income

X_2 = Spending Score

Set $K = 3$

Initialize 3 centroids

REPEAT

 Calculate distance from each customer
 to each centroid

 Assign customer to nearest centroid

 Calculate new centroid for each cluster

UNTIL centroids stop changing

Display Cluster 1, Cluster 2 and Cluster 3

STOP

Code

```
fds co5 exp3.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python
File Edit Format Run Options Window Help
import pandas as pd
from sklearn.cluster import KMeans

data = pd.DataFrame({
    "Income": [20, 25, 30, 70, 75, 80, 40, 45, 50],
    "SpendingScore": [80, 75, 85, 20, 25, 15, 50, 55, 45]
})

X = data[["Income", "SpendingScore"]]

model = KMeans(n_clusters=3, random_state=42, n_init=10)

data["Cluster"] = model.fit_predict(X)

print(data)

print("\nCentroids:")
print(model.cluster_centers_)

IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:0
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Progra
o5 exp3.py
   Income  SpendingScore  Cluster
0      20             80         2
1      25             75         2
2      30             85         2
3      70             20         0
4      75             25         0
5      80             15         0
6      40             50         1
7      45             55         1
8      50             45         1

Centroids:
[[75. 20.]
 [45. 50.]
 [25. 80.]]
>>> |
```

Result

Thus, customers were successfully grouped into three clusters based on annual income and spending score.

Experiment 4 – K-Means Using Elbow Method

Aim

To design pseudocode and implement the Elbow Method to determine a suitable number of clusters for K-Means clustering.

Pseudocode

START

Load dataset

FOR K = 1 to 10:

 Apply K-Means clustering

 Calculate WCSS

 Store WCSS value

Plot K against WCSS

Find the elbow point

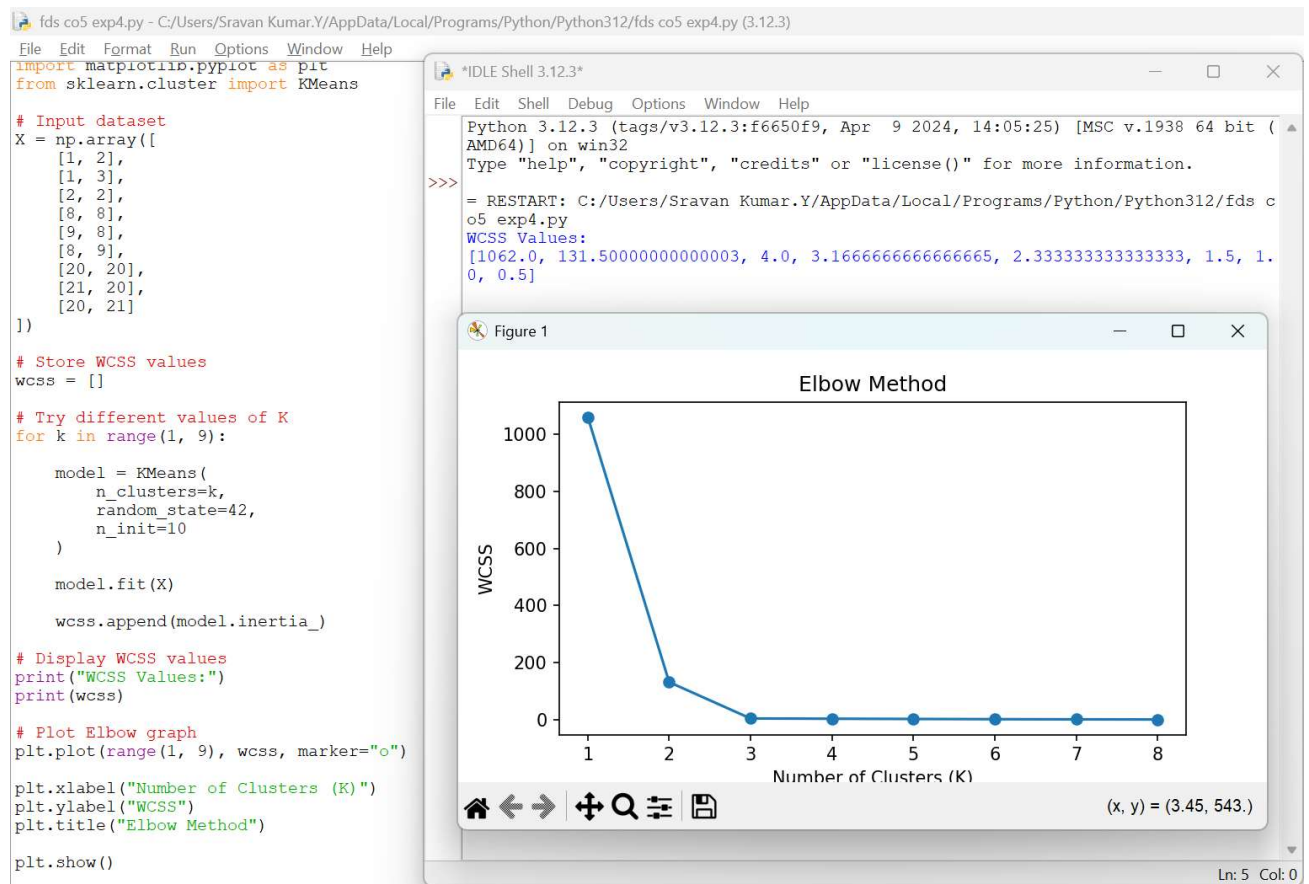
Select K at the elbow point

Apply K-Means using selected K

Display final clusters

STOP

Code



Result

Thus, the Elbow Method was successfully used to determine a suitable value of K for K-Means clustering.

Experiment 5 – Evaluation Metrics

Aim

To design pseudocode and implement the calculation of TP, TN, FP, FN, accuracy, precision, recall, and F1-score for a binary classification model.

Pseudocode

START

Input actual labels

Input predicted labels

Initialize:

TP = 0

TN = 0

FP = 0

FN = 0

FOR each prediction:

IF Actual = 1 AND Predicted = 1:

TP = TP + 1

ELSE IF Actual = 0 AND Predicted = 0:

TN = TN + 1

ELSE IF Actual = 0 AND Predicted = 1:

FP = FP + 1

ELSE IF Actual = 1 AND Predicted = 0:

FN = FN + 1

Calculate Accuracy

Calculate Precision

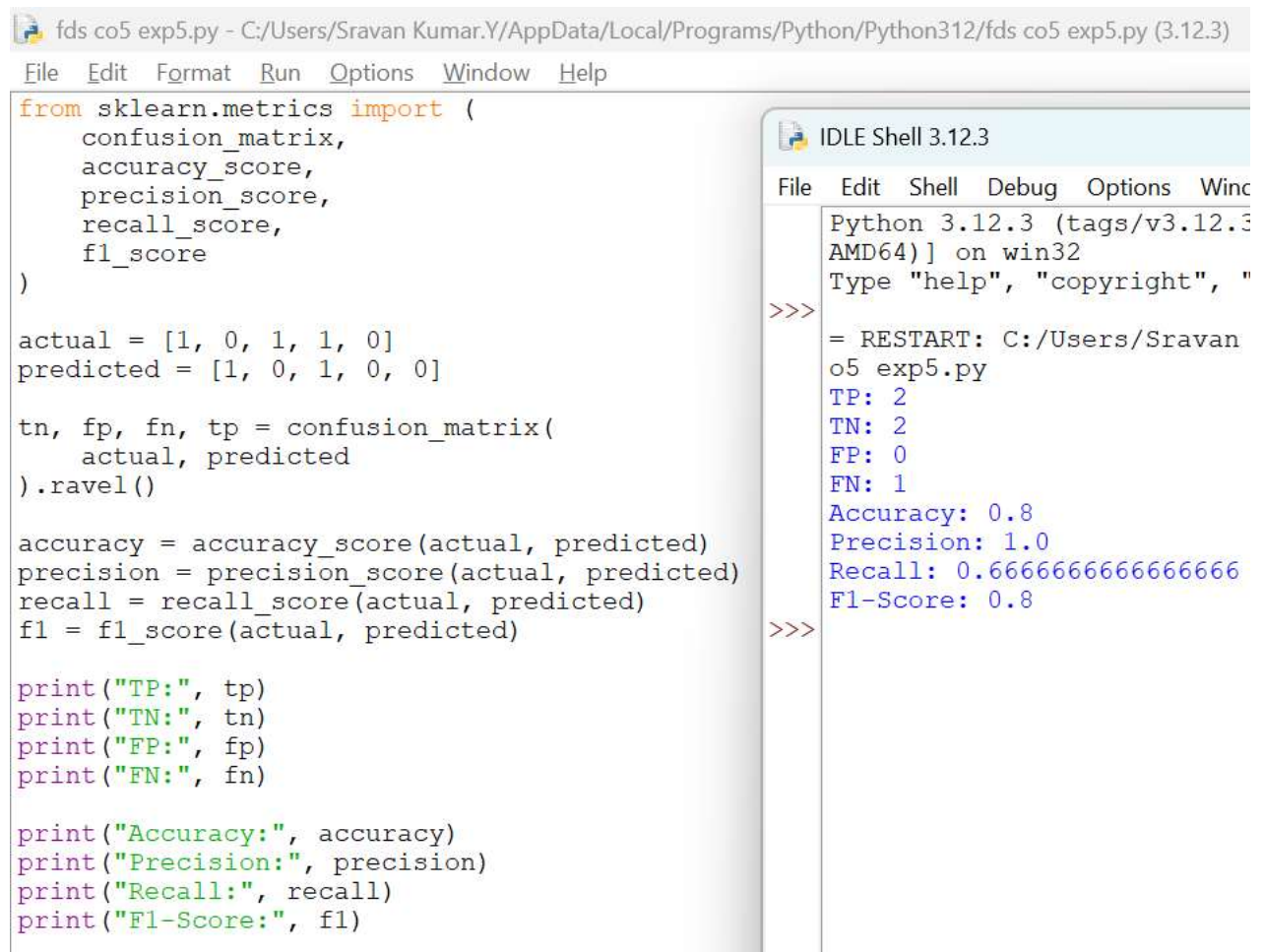
Calculate Recall

Calculate F1-score

Display results

STOP

Code



```
fds co5 exp5.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds co5 exp5.py (3.12.3)
File Edit Format Run Options Window Help

from sklearn.metrics import (
    confusion_matrix,
    accuracy_score,
    precision_score,
    recall_score,
    f1_score
)

actual = [1, 0, 1, 1, 0]
predicted = [1, 0, 1, 0, 0]

tn, fp, fn, tp = confusion_matrix(
    actual, predicted
).ravel()

accuracy = accuracy_score(actual, predicted)
precision = precision_score(actual, predicted)
recall = recall_score(actual, predicted)
f1 = f1_score(actual, predicted)

print("TP:", tp)
print("TN:", tn)
print("FP:", fp)
print("FN:", fn)

print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
print("F1-Score:", f1)

IDLE Shell 3.12.3
File Edit Shell Debug Options Window
Python 3.12.3 (tags/v3.12.3:33608c0d, Apr 19 2024, [AMD64]) on win32
Type "help", "copyright", "credits() or "license()" for more
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds co5 exp5.py
TP: 2
TN: 2
FP: 0
FN: 1
Accuracy: 0.8
Precision: 1.0
Recall: 0.6666666666666666
F1-Score: 0.8
>>>
```

Result

Thus, the classification model was successfully evaluated using confusion matrix values and standard evaluation metrics.

Experiment 6 – Model Evaluation and Comparison

Aim

To design pseudocode and implement a comparison of kNN, Decision Tree, Logistic Regression, and K-Means models using appropriate evaluation measures.

Pseudocode

START

Load dataset

Preprocess dataset

Split data into training and testing sets

Train kNN model

Evaluate kNN

Train Decision Tree model

Evaluate Decision Tree

Train Logistic Regression model

Evaluate Logistic Regression

Apply K-Means clustering

Evaluate K-Means

Compare all model results

Select the best suitable model

Display selected model

STOP

Code

```
fds co5 exp6.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds co5 exp6.py (3.12.3)
File Edit Format Run Options Window Help
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.cluster import KMeans
from sklearn.metrics import accuracy_score, silhouette_score

# Load dataset
data = load_iris()

X = data.data
y = data.target

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# kNN
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
knn_pred = knn.predict(X_test)
knn_accuracy = accuracy_score(y_test, knn_pred)

# Decision Tree
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)
dt_pred = dt.predict(X_test)
dt_accuracy = accuracy_score(y_test, dt_pred)

# Logistic Regression
lr = LogisticRegression(max_iter=200)
lr.fit(X_train, y_train)
lr_pred = lr.predict(X_test)
lr_accuracy = accuracy_score(y_test, lr_pred)

# K-Means
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
clusters = kmeans.fit_predict(X)
silhouette = silhouette_score(X, clusters)

# Display results
print("kNN Accuracy:", knn_accuracy)
print("Decision Tree Accuracy:", dt_accuracy)
print("Logistic Regression Accuracy:", lr_accuracy)
print("K-Means Silhouette Score:", silhouette)
```

```
IDLE Shell 3.12.3
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Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024,
AMD64) on win32
Type "help", "copyright", "credits" or "license()"
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/I
o5 exp6.py
kNN Accuracy: 1.0
Decision Tree Accuracy: 1.0
Logistic Regression Accuracy: 1.0
K-Means Silhouette Score: 0.5528190123564095
>>>
```

Result

Thus, kNN, Decision Tree, Logistic Regression, and K-Means were successfully applied and evaluated using appropriate performance measures, and their results were compared to identify a suitable model.